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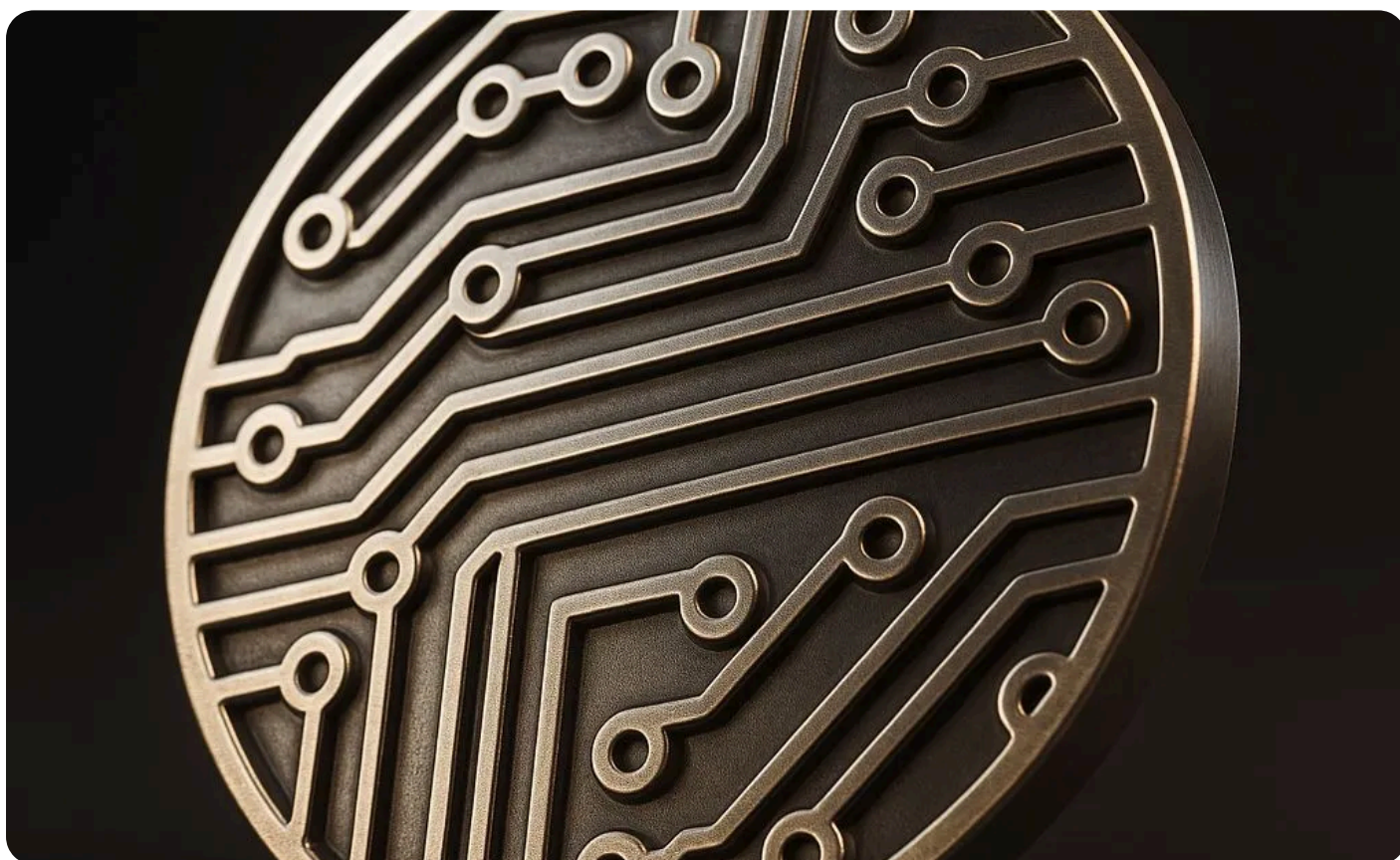
CORE RESEARCH

PTFE Pushing Copper Further

// Multi-Vertical Note

3 MINUTES

By Tony Chien and Wega Chu



Executive Summary:

- Nvidia aims to implement PTFE for the NVSwitch board of Rubin Ultra, however, final design and adoption are to be finalized depending on the readiness of PTFE and PCB vendors.
- Nvidia's effort with PTFE has been ongoing for more than a year with Shengyi leading the development. Recently, WUS has successfully produced a PTFE PCB sample.
- The current solution is a hybrid approach using PTFE layer as core and M9 grade prepreg. This solution combines the signal performance advantage of PTFE and the mechanical support of M9 grade CCL, allowing for an easier manufacturing process than a standalone PTFE solution.
- EMC is partnering with TaiFlex as both needs each other for the hybrid solution given their expertise in the M9 grade glass-cloth based CCL and PTFE based CCL respectively.

Back in GTC 2025, we mentioned Nvidia's intention to qualify and use PTFE as an alternative solution to the current glass fiber-based CCL, in order to extend the lifecycle of copper for scale up network application. With Kyber, the product which we assumed will adopt PTFE contents, delayed, Nvidia is still looking to incorporate PTFE materials into its Rubin Ultra Oberon deployment. As it stands, Nvidia aims to use PTFE for the Portia (NVSwitch 7) board under a hybrid solution with M9 grade glass cloth CCL. However, the design is not finalized, and adoption depends on the maturity and readiness of the technology and PCB vendors by the time Rubin Ultra ramps.

Crash Course on Resin and PTFE

To understand why Nvidia wants to adopt PTFE, we must first understand why PTFE provides better signal performance compared to glass fiber-based CCL. The key to this is all about the resin and its molecular structure.

For the resin, good electrical performance requires low polarity and a symmetrical molecular structure. The history of resin development follows this requirement from epoxy, PPO, the introduction of hydrocarbon, and finally to PTFE.

The material used in FR-4 to M4 CCL was epoxy resin, which contains many epoxide groups surrounding its main chemical structure. As the industry moved toward low-loss materials, CCL makers began developing polyphenylene oxide (PPO), also known as polyphenylene ether (PPE). Its benzene structure — the familiar hexagonal “turtle-shell” ring (C_6H_6) — is a flat, six-carbon ring, which gives this material a more symmetrical molecular structure.

Further down the road, the ultra-low-loss M7 generation began incorporating hydrocarbons blended with benzene rings. Hydrocarbon has a backbone made almost entirely of C-C and C-H bonds. The low polarity of C-H bonds in the polyolefin backbone, combined with the molecular chain’s zigzag conformation, gives the resin excellent dielectric properties. By M9 grade CCL, the resin system had become mostly hydrocarbon based.

Why is PTFE the only solution for next level dielectric performance?

PTFE (Polytetrafluoroethylene) has a symmetric structure, where fluorine atoms are distributed evenly around the carbon backbone, so the molecule has essentially no net polarity, this is what gives PTFE its very low dielectric constant and dielectric loss ($D_k = 2.1$, $D_f = 0.0003$ at 10 GHz). This makes it the lowest-loss common polymer used in CCL, beating both PPO/PPE and standard hydrocarbon resins.

Fluorine atoms shield the carbon backbone far more effectively than hydrogen, and the C-F bond energy reaches about 460.5 kJ/mol, much higher than C-H or C-C bonds, which is why PTFE has outstanding chemical stability and corrosion resistance. This bond-strength is the real source of its “stability” — it resists thermal degradation, oxidation, and chemical attack far better than hydrocarbon or PPO chains.

CCL Grades — Electrical Properties						
		M6	M7	M8	M9	PTFE
Dk @ 1 GHz		3.34-3.71	3.37-3.63	Not tested	Not tested	Not tested
Dk @ 10 GHz		3.61	3.35-3.61	3.13-3.27	3.0-3.1	2.1-2.2
Df @ 1 GHz		0.002	0.0015-0.002	Not tested	Not tested	Not tested
Df @ 10 GHz		0.004	0.002-0.003	0.0016-0.0018	0.00046-0.00050	<0.0004

Sources: SemiAnalysis

Challenges with PTFE

Across these materials, the shift toward hydrocarbon and fluoropolymer (PTFE) resins comes with a trade-off: molecular mass increases while mobility decreases. This creates challenges during circuit manufacturing. After drilling, a resin smear could be left behind and would be difficult to remove. Put simply, the lower the material's polarity, the harder it is to process in PCB manufacturing.

Moreover, Epoxy, PPO/PPE, and hydrocarbon resins are all thermosetting resins, meaning that once heated to a certain temperature, they undergo an irreversible cross-linking reaction. PTFE, by contrast, is thermoplastic and does not undergo this irreversible cross-linking, meaning it will be difficult to withstand multiple lamination/pressing and PCBA cycles.

This difference also makes PTFE notoriously difficult to process in manufacturing. Being slippery and soft, and lacking mechanical strength, it is prone to burring and poor hole-wall quality during drilling and harder to maintain dimensional stability.

What is the current solution for PTFE?

The current solution is a hybrid approach using PTFE as core layers and M9 grade prepreg. Since the board is not made entirely of PTFE layers, the glass cloth layers provide mechanical strength. In other words, this design eases the manufacturing process.

The PTFE participants for AI servers are highly concentrated, with only a few key players working alongside Nvidia. Shengyi Technology (600183 SS) has been closely developing PTFE with Nvidia for more than 12 months, putting it ahead of others in this field. Shengyi has already delivered PTFE materials to multiple PCB vendors for sampling. While the effort has been muted for the last year, recently, WUS has successfully delivered a PTFE-material PCB sample. Meanwhile, EMC (2383 TT) has partnered with TaiFlex (8039 TT). The

partnership leverages EMC's experience in the Megtron-grade CCL/prepreg and TaiFlex's specialty in PFA (Perfluoroalkoxy Alkane), a high-performance fluoropolymer that combines the extreme chemical resistance and thermal stability of PTFE with the melt-processability of conventional thermoplastics, to provide a hybrid solution to Nvidia.

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